

SENTINELX

Autonomous AI Threat Hunting Platform

“See every threat. Miss nothing. Respond before impact.”

Hackathon	Problem Statement	Team
Forge the Future Hackathon 2026	Threat Hunting at Scale	[Team Name Placeholder]

SentinelX is an autonomous, multi-agent AI platform built on Elastic Search AI and AWS that continuously hunts, correlates, and explains security threats across enterprise environments — cutting investigation time from hours to minutes.

Executive Summary

Enterprise SOCs ingest millions of log events daily across endpoints, cloud, network, and identity systems — far more than human analysts can review. Current SOCs rely on static rules and manual triage, producing severe alert fatigue, slow investigations, and analysts who spend more time searching for context than responding to real threats. As a result, Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) stay dangerously high while sophisticated attackers hide inside the noise.

SentinelX solves this with a team of coordinated AI agents that continuously monitor Elastic-indexed telemetry, autonomously investigate anomalies, reconstruct full attack chains, and generate explainable, analyst-ready findings with recommended remediation. Built on Elastic Search AI for retrieval and AWS Bedrock for reasoning, SentinelX turns threat hunting from a reactive, manual chore into a proactive, always-on capability — reducing analyst workload, cutting response time, and closing the detection gap that manual SOC operations cannot.

Problem Statement

- Volume overload: enterprises generate millions of security log events per day across cloud, endpoint, and network layers.
- Manual investigation is slow: analysts manually pivot across disconnected tools to piece together a single incident.
- Alert fatigue: high false-positive rates cause critical alerts to be ignored or delayed.
- Hidden threats persist: low-and-slow, multi-stage attacks evade static, rule-based detection.
- High MTTD: real threats often go unnoticed for days.
- High MTTR: once found, response is delayed by manual root-cause analysis and reporting.

Bottom line: SOC teams are drowning in data but starving for insight.

Proposed Solution

SentinelX is an autonomous AI-powered threat hunting platform that continuously monitors security logs, detects anomalies, investigates attacks, correlates related events, performs semantic search across historical incidents, reconstructs full attack chains, and produces explainable AI reasoning with recommended remediation — automatically generating investigation-ready reports for SOC analysts.

The platform is built on Elastic Search AI for real-time ingestion, semantic and vector search across security telemetry, and AWS Bedrock for LLM-driven reasoning, correlation, and natural-language explanation — giving enterprises a scalable, explainable, and production-ready alternative to manual threat hunting.

Core Features

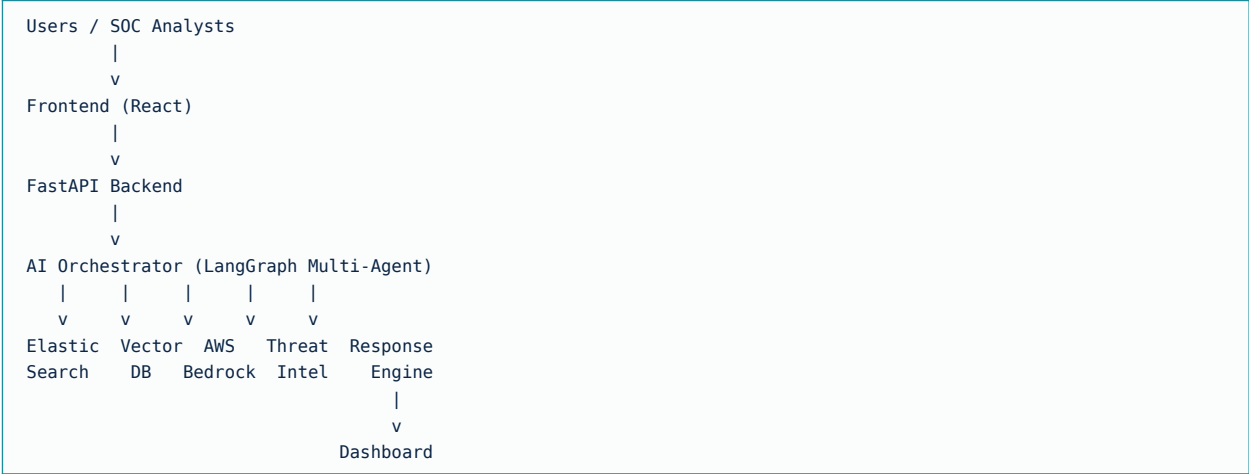
Feature	Description
Continuous Log Monitoring	Real-time ingestion and indexing of security telemetry via Elastic.
Anomaly Detection	ML-driven detection of deviations from established behavioral baselines.
Semantic Threat Search	Natural-language querying across historical logs and past incidents.
Automated Attack Correlation	Links related events across systems into a single incident timeline.
Attack Chain Reconstruction	Rebuilds the full sequence of an attack, from entry to impact.
Explainable AI Reasoning	Every finding includes a human-readable rationale, not a black-box score.
Remediation Recommendations	Actionable, prioritized response steps for each confirmed threat.
Autonomous Report Generation	Analyst-ready incident reports generated without manual write-up.
Multi-Agent Orchestration	Specialized agents collaborate on detection, analysis, and response.
Enterprise-Grade Scalability	Cloud-native design handles enterprise log volume and concurrency.

AI Agent Workflow

1. **Threat Detection Agent** — Continuously scans indexed logs for behavioral anomalies and known attack signatures.
2. **Correlation Agent** — Links related alerts and events across systems into a unified incident context.
3. **Threat Intelligence Agent** — Enriches findings with external threat intel feeds and known adversary tactics.
4. **Root Cause Analysis Agent** — Traces the incident back to its origin and reconstructs the attack chain.
5. **Response Agent** — Recommends and, where authorized, triggers containment and remediation actions.
6. **Report Generation Agent** — Synthesizes findings into a clear, explainable report for SOC review.

High-Level Architecture

Users → Frontend (React) → FastAPI Backend → AI Orchestrator (LangGraph) → [Elastic Search | Vector DB | AWS Bedrock | Threat Intel APIs] → Response Engine → Dashboard



Technology Stack

Layer	Technology
Frontend	React, TypeScript, TailwindCSS
Backend	FastAPI (Python), REST/WebSocket APIs
AI / Orchestration	LangGraph, multi-agent reasoning, prompt orchestration
Elastic	Elastic Search AI, ELSER semantic search, Kibana
AWS	AWS Bedrock, Lambda, S3, CloudWatch
Database	Vector database (embeddings), PostgreSQL (metadata)
DevOps	Docker, GitHub Actions CI/CD, Terraform

Expected Outcomes

Metric	Impact
Investigation Time	↓ 70%
False Positives	↓ 50%
Mean Time to Respond (MTTR)	↓ 60%
Monitoring Coverage	24×7 autonomous
Findings Transparency	Fully explainable AI
Deployment Readiness	Enterprise-scale

Future Scope

- Predictive Threat Hunting — forecast likely attack paths before exploitation.
- Zero Trust Integration — continuous identity and access verification signals.
- Multi-Cloud Security — unified hunting across AWS, Azure, and GCP.
- SOAR Integration — automated playbook-driven response orchestration.
- Autonomous Patch Recommendation — AI-suggested prioritized patching guidance.

Why SentinelX?

SentinelX is not another dashboard or alert generator — it's an autonomous SOC teammate. By combining Elastic's real-time search and semantic retrieval with AWS Bedrock's reasoning capability, SentinelX delivers explainable, production-ready threat hunting that scales with enterprise complexity instead of drowning in it. It directly reflects Elastic's vision for AI that is transparent, trustworthy, and built for real operational impact — making SentinelX a natural fit for enterprises ready to move from reactive security to autonomous defense.